

TV Show / Movie Genre Prediction

Model Evaluation & Explainability Report

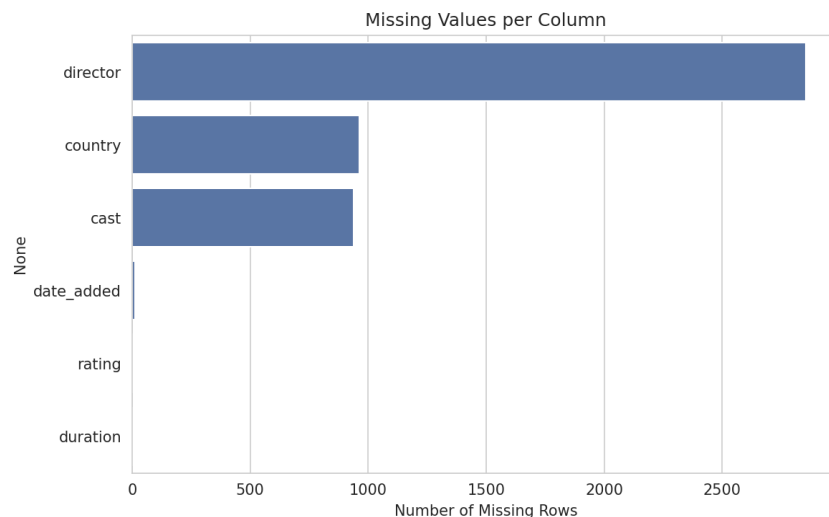
1. Objective & Dataset

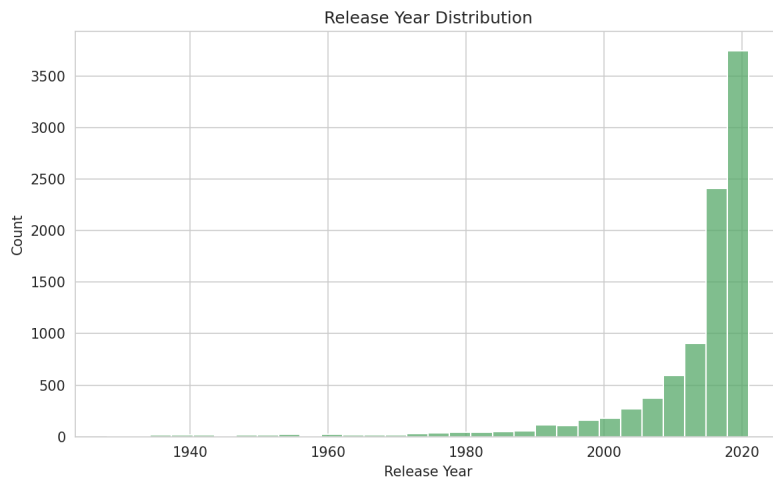
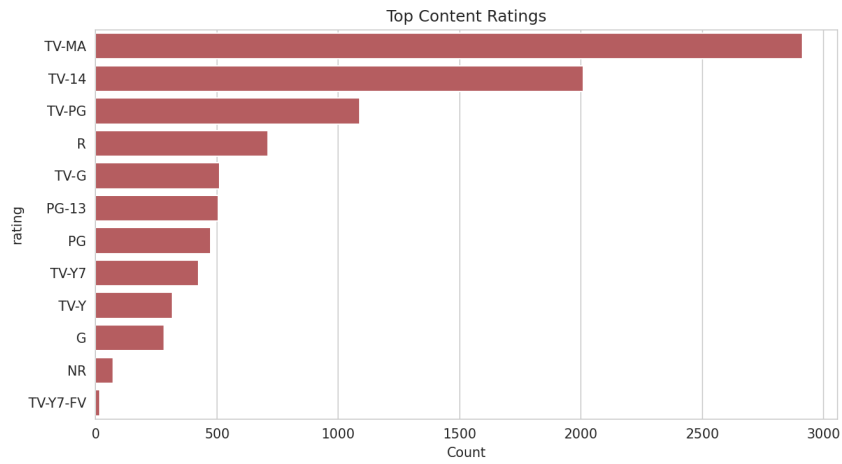
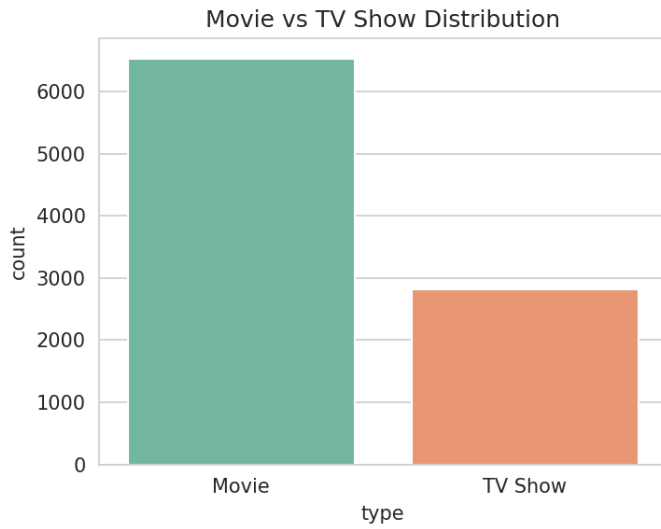
The goal is to predict the genre of a Netflix / Disney title (Movie or TV Show) from its metadata and description. The dataset contains 9,338 titles with fields including type, director, cast, country, release year, content rating, duration, a free-text description, and the raw multi-label listed_in genre tags.

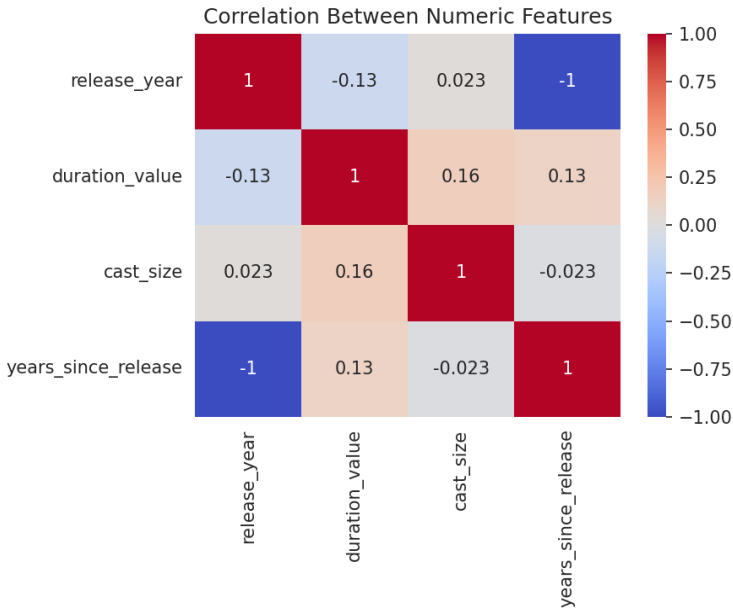
2. Data Understanding & EDA

Key observations from exploratory analysis:

- The dataset is dominated by Movies (6,528) over TV Shows (2,810), and by the Netflix platform (7,926) over Disney (1,412).
- Director (2,855), cast (938), and country (962) have substantial missing values; these were imputed or absorbed into an 'Unknown' category rather than dropped.
- Release years are heavily skewed toward recent content (median 2017), reflecting streaming catalogs' recency bias.
- TV-MA and TV-14 are the most common ratings, indicating a mature-audience-skewed catalog.







3. Target Construction

The raw `listed_in` column is multi-label with 84 granular tags (e.g. 'TV Dramas', 'Dramas', 'Action-Adventure', 'Action & Adventure' largely duplicated concepts split by Movie/TV naming). To create a learnable single-label target, the first-listed genre per title was mapped to one of 13 broad genre classes via keyword matching. Classes with fewer than 40 samples (Romance, Sci-Fi & Fantasy) were dropped, leaving 11 target classes used for modeling.



Comedy and Drama are the largest classes; Reality & Talk Show is the smallest at 112 samples. This moderate imbalance motivated the use of `class_weight='balanced'` in both models.

4. Preprocessing & Features

Features used: `release_year`, `years_since_release`, `duration_value` (minutes or number of seasons), and `cast_size` as numeric features (median-imputed, scaled); `type`, `rating`, `platform`, and `country_main` as categorical features (most-frequent-imputed, one-hot encoded); and `description` as free text (TF-IDF, 300 terms, English stop words removed). All steps are combined into a single scikit-learn `ColumnTransformer` so the exact same preprocessing is applied consistently at train and inference time. Data was split 80/20 with stratification on the target (`seed=42`).

5. Model Development

Two traditional ML models were trained on identical preprocessed features for a fair comparison:

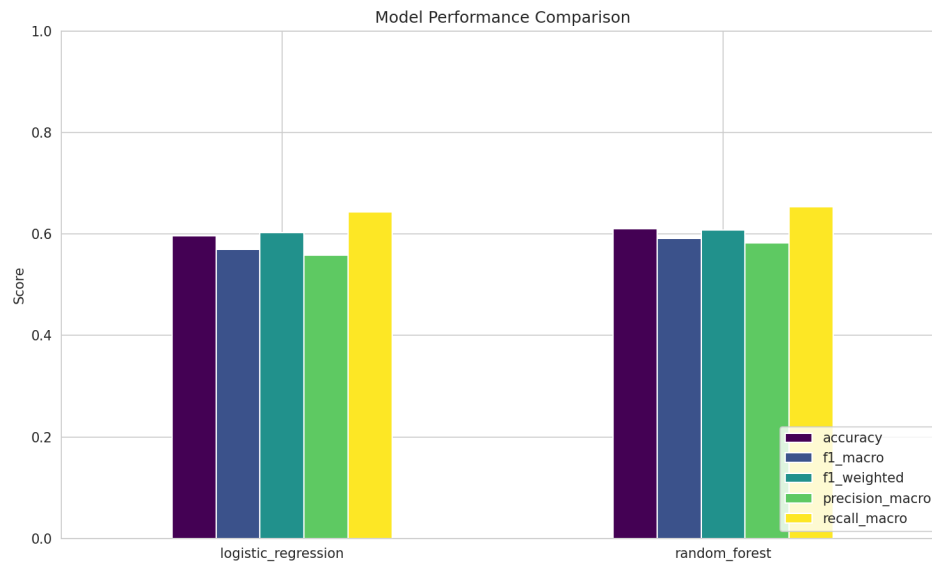
- Logistic Regression — linear model, `class_weight='balanced'`, `max_iter=2000`. Fast, interpretable via coefficients.

- Random Forest — 300 trees, `class_weight='balanced'`. Captures non-linear interactions between features.

6. Evaluation Results

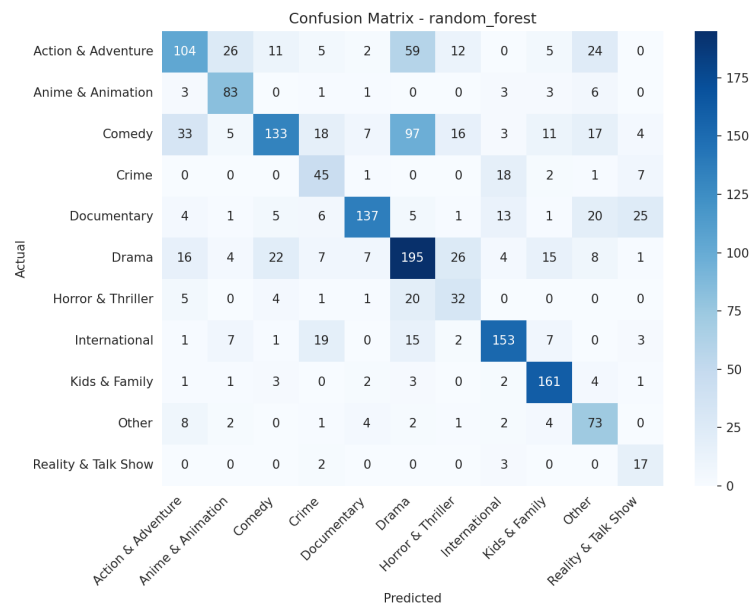
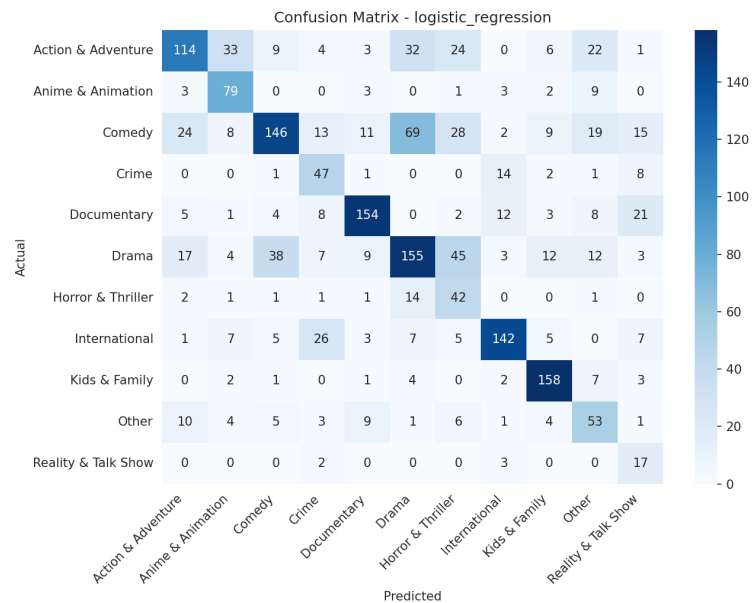
Metrics on the held-out 20% test set (1,857 titles):

Model	Accuracy	F1 (macro)	F1 (weighted)	Precision (macro)	Recall (macro)	ROC-AUC (macro OvR)
Logistic Regression	0.596	0.570	0.603	0.558	0.644	0.929
Random Forest	0.610	0.592	0.608	0.582	0.653	0.938



Random Forest slightly outperforms Logistic Regression across every metric, most notably accuracy (61.0% vs 59.6%) and ROC-AUC (0.938 vs 0.929). Both models achieve strong ROC-AUC scores (>0.92), indicating they rank genre probabilities well even though top-1 accuracy is more modest expected given 11 overlapping genre classes derived from noisy, multi-label source tags.

Confusion Matrices



Both models perform best on Kids & Family, Anime & Animation, International, and Documentary genres with distinctive vocabulary and metadata patterns. Both struggle most to separate Drama, Comedy, and Action & Adventure from each other, and Horror

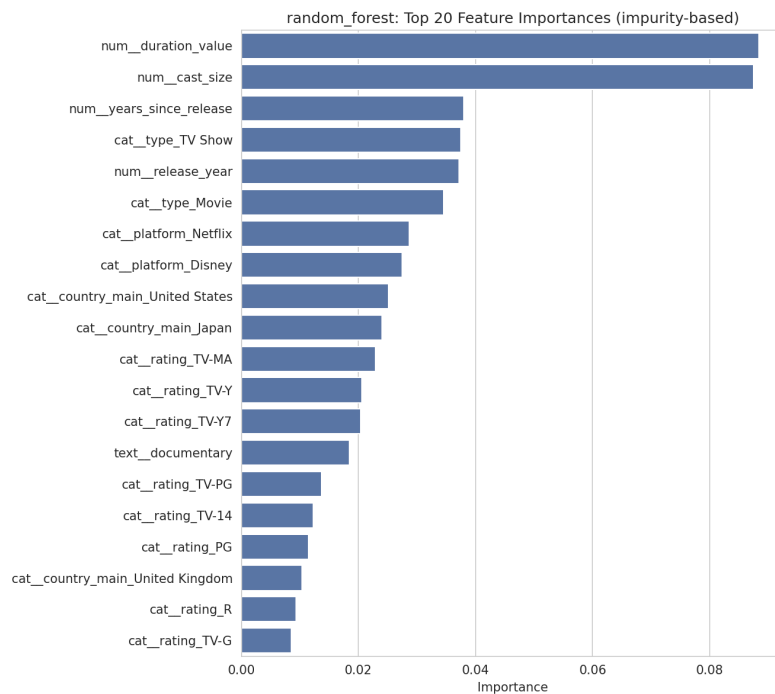
& Thriller / Reality & Talk Show suffer from small sample sizes and label overlap with Drama/Documentary respectively.

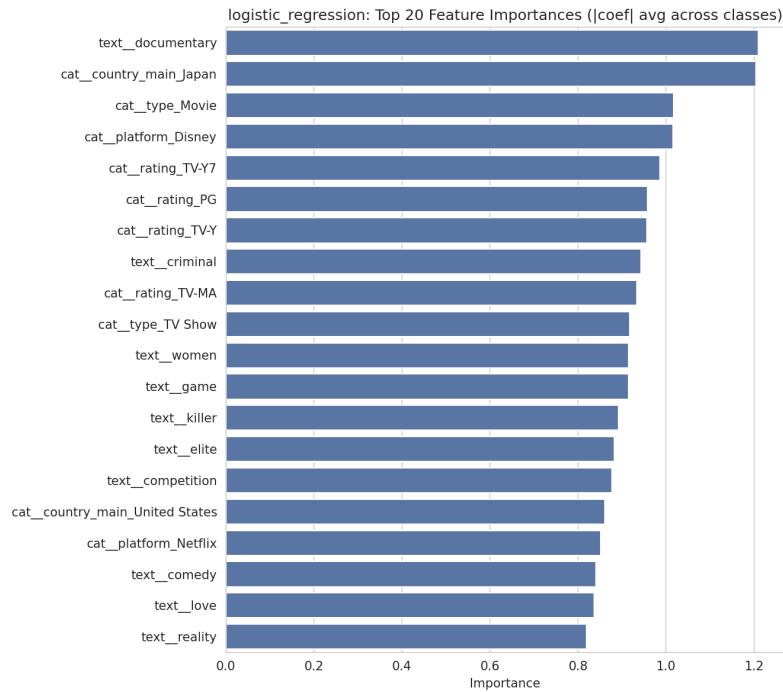
7. Explainability

Three complementary explainability views were produced for each model:

7.1 Built-in Feature Importance

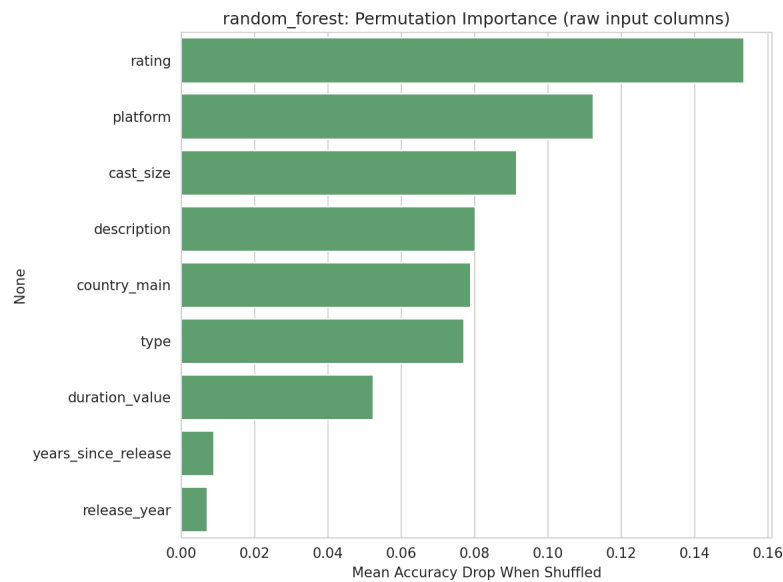
Random Forest impurity-based importance and Logistic Regression average absolute coefficient magnitude, both computed on the encoded/TF-IDF feature space.

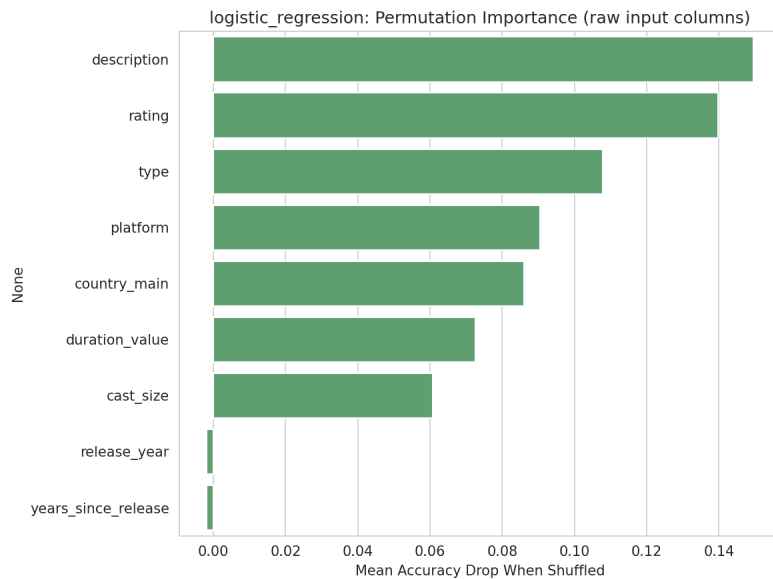




7.2 Permutation Importance (raw input columns)

A model-agnostic view: each raw input column is shuffled and the resulting drop in test accuracy is measured. This is the most interpretable importance view since it operates on the original columns rather than encoded features.

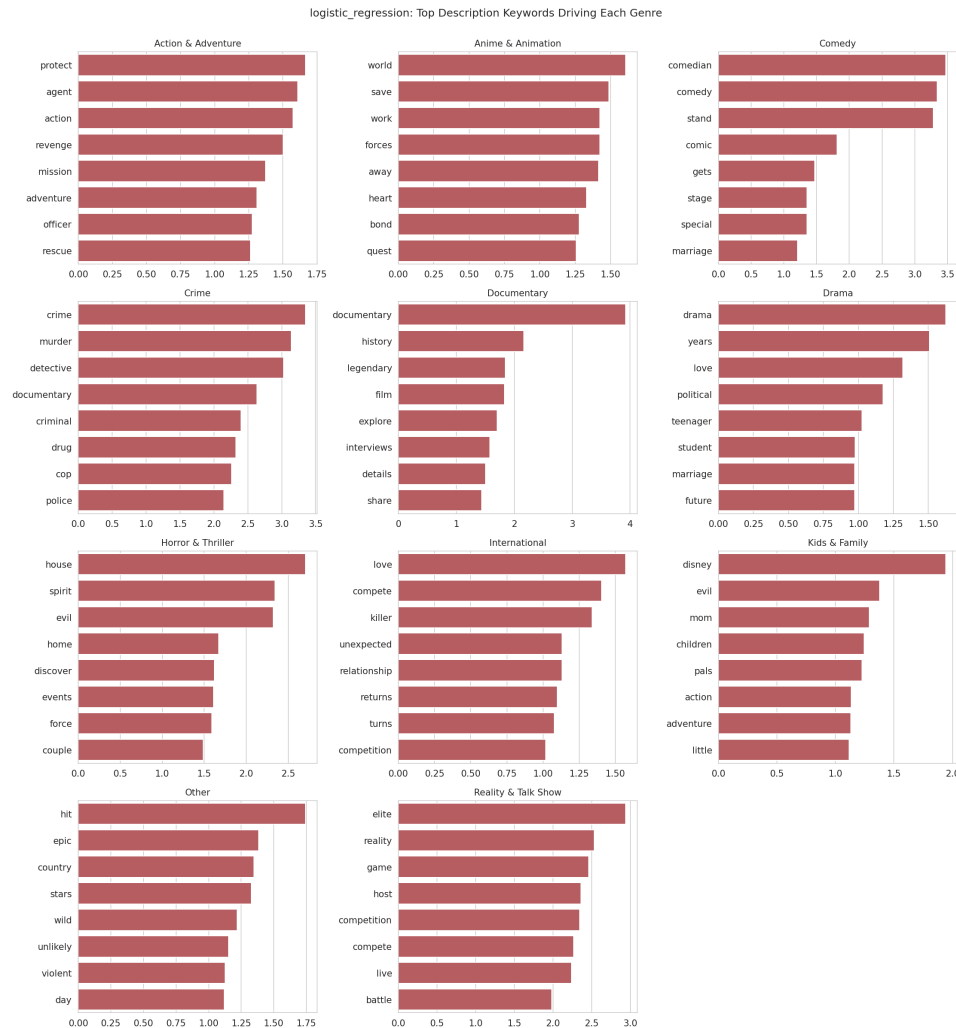




Content rating, platform, and the text description are consistently the strongest predictors across both models. This makes intuitive sense: rating correlates with genre (e.g. TV-Y is almost always Kids & Family), platform reflects each service's differing content mix, and the description contains genre-revealing keywords directly.

7.3 Top Description Keywords per Genre (Logistic Regression)

For the linear model, the TF-IDF terms with the strongest positive coefficient for each genre class reveal which words in a description push a prediction toward that genre e.g. terms like 'friends', 'family' driving Kids & Family, or 'crime', 'detective' driving Crime.



8. Recommendation

We recommend the Random Forest model for this task. It outperforms Logistic Regression on every evaluation metric (accuracy, macro/weighted F1, precision, recall, and ROC-AUC), while requiring no additional feature engineering to capture non-linear relationships and interactions between categorical metadata (e.g. rating x platform x duration) that a linear model cannot represent directly.

That said, Logistic Regression remains a strong, more interpretable baseline: its coefficients map directly and transparently to genre-specific keywords, which is valuable if model transparency is a priority over the

last few points of accuracy. In a production setting, Random Forest is the better default model, with Logistic Regression retained as an interpretable audit/reference model.

Both models achieve ROC-AUC scores above 0.92 despite moderate top-1 accuracy (~60%), reflecting the inherent difficulty of the task: the target genres were derived from multi-label, overlapping source tags (a title tagged 'Crime, Drama, Thriller' only has one 'correct' primary label), so some ceiling on achievable accuracy is expected from the label construction itself, not just model capacity.

9. Reproducing These Results

See README.md for full setup instructions. In summary:

- `pip install -r requirements.txt`
- `python src/train.py --data_path data/tv-shows.csv --model all --seed 42`
- `python src/evaluate.py --model_path all --data_path data/tv-shows.csv`
- `python src/explain.py --output_dir outputs`